

wiring_instructions

Semgrep wiring — apply instructions (conductor / operator)

Constitution: §11.4.166 (Universal Semgrep static analysis mandate). **Date:** 2026-06-22 **Working dir:** /Volumes/T7/Projects/helix_code **Evidence basis:** docs/research/semgrep_onboarding_20260622/findings.md

The onboarding agent created only DISJOINT new files. It did **NOT** touch .mcp.json, ~/.zshrc, ~/.bashrc, or the live .git/hooks/pre-commit (meta-git race avoidance + live-hook safety, §11.4.6). All four wiring points below are for the conductor/operator to apply.

Scripts are inherited **by reference** from the constitution submodule (§11.4.28 / §11.4.35 / §11.4.74) — wire, never copy.

Files the agent created (DISJOINT, safe to commit)

File	Purpose
.docs_chain/contexts/semgrep_status.yaml	docs_chain context for the semgrep Status ledger (§11.4.166(6) / §11.4.106)
docs/semgrep/Status.md	consumer-side semgrep event ledger (SOURCE for the docs_chain context)
docs/research/semgrep_wiring_20260622/precommit_wiring.md	exact pre-commit hook change
docs/research/semgrep_wiring_20260622/mcp_entry.json	exact .mcp.json “semgrep” entry
docs/research/semgrep_wiring_20260622/wiring_instructions.md	this file

(1) docs_chain context — DONE (file created)

.docs_chain/contexts/semgrep_status.yaml registers a semgrep_status context: docs/semgrep/Status.md (markdown SOURCE) → Status.html (md-to-html / pandoc) → Status.pdf (html-to-pdf / weasyprint). Schema matches the existing contexts (governance, fixed, helixcode).

Generate the exports:

```
cd /Volumes/T7/Projects/helix_code
docs_chain sync    --root . semgrep_status    # regenerate
                  Status.html + Status.pdf
docs_chain verify --root . semgrep_status    # pre-build gate
                  (must pass)
```

(If docs_chain resolves to the engine per §11.4.106; the constitution exposes it. The .html/.pdf siblings are gitignored-derivatives-regenerated, not hand-authored.)

(2) Pre-commit hook — apply per precommit_wiring.md

The live .git/hooks/pre-commit is the guard-forbidden-commands.sh guard (§11.4.109), with NO semgrep reference. Wire semgrep **additively** (do not weaken the guard). RECOMMENDED (Option A in precommit_wiring.md): replace the hook's single trailing exit 0 with:

```
_SEMGREP_HOOK="$(git rev-parse --show-toplevel)/constitution/
                  scripts/hooks/semgrep_precommit.sh"
if [ -x "$_SEMGREP_HOOK" ]; then
    "$_SEMGREP_HOOK" || exit 1
fi

exit 0
```

This runs semgrep scan --config auto --error on staged source files and blocks the commit (exit 1) on any finding; graceful-degrades when semgrep is absent. Full detail + Option B (via scripts/install_git_hooks.sh) + an inline one-liner alternative in precommit_wiring.md.

(3) .mcp.json “semgrep” server — apply per mcp_entry.json

Add the semgrep key from mcp_entry.json into the existing mcpServers object in /Volumes/T7/Projects/helix_code/.mcp.json (sibling to codegraph, media-validator, open-design). Do **not** replace the file — insert the one key. Verified for this host: uvx is present at /opt/homebrew/bin/uvx (FACT), so uvx semgrep-mcp is launchable; uvx fetches semgrep-mcp on demand. The entry pins SEMGREP_PATH=/opt/homebrew/bin/semgrep (the working 1.167.0 binary confirmed this session).

After insertion, validate the merged file:

```
python3 -m json.tool /Volumes/T7/Projects/helix_code/.mcp.json >/dev/null && echo "VALID JSON"
```

Then restart the agent session so the new MCP server loads, and confirm semgrep appears in the live MCP server list (install exit 0 != loadable — confirm by observing it live, §11.4.102(C)).

Note: this environment also exposes the marketplace semgrep MCP plugin as mcp__plugin_semgrep_semgrep__*. The .mcp.json uvx semgrep-mcp entry is the project-pinned, decoupled form (§11.4.28) required by §11.4.166 so the wiring is reproducible from the checkout regardless of marketplace plugins.

(4) Shell rc PATH line — operator applies

Add to ~/.zshrc (and/or ~/.bashrc) so semgrep is on PATH in every shell (§11.4.166(2)):

```
source /Volumes/T7/Projects/helix_code/constitution/scripts/semgrep/semgrep_path.sh
```

(Currently moot for interactive use since semgrep is already on PATH via Homebrew, but the rc source line is required by §11.4.166(2) for all-shells/all-users availability and is harmless — semgrep_path.sh only prepends common bin dirs, dedup-guarded, and does not require semgrep pre-installed.)

Post-apply verification (§11.4.166 / §107)

```
cd /Volumes/T7/Projects/helix_code
# validate (anti-bluff JSON-with-results probe)
```

```
sh constitution/scripts/semgrep/semgrep_validate.sh
# CI integration test (known-vuln detection, evidence under qa-
  results/)
sh constitution/scripts/semgrep/semgrep_ci_test.sh
# docs_chain export gate
docs_chain verify --root . semgrep_status
```

Sources verified 2026-06-22

- constitution/scripts/semgrep/
 {semgrep_path,semgrep_validate,semgrep_setup,semgrep_ci_test}.sh
- constitution/scripts/hooks/semgrep_precommit.sh
- .git/hooks/pre-commit, .mcp.json, .docs_chain/contexts/*.yaml
 (read this session)
- host probe: uvx=/opt/homebrew/bin/uvx, semgrep=1.167.0 (this session)
- Constitution §11.4.166; §11.4.28/§11.4.35/§11.4.74/§11.4.102(C)/§11.4.106